4.5 Graphing and Interpreting Key Features of Lines

Lesson Introduction

Benchmark	MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.	Learning Target	• I can graph a line and interpret its key features.

Benchmark Coherence	Building On	Working Toward
	MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.	MA.912.F.1.7 Compare key features of two functions each represented algebraically, graphically, in tables or written descriptions.

Essential Questions	• Why is context important when interpreting domain and range? • Which intercept represents the solution to an equation or function? • How does the context of the problem influence or inform the key features?
Lesson Narrative	Students will practice graphing lines from both a table of values and an equation before further interpreting the key features in the next lesson. They will begin to make conjectures about how to interpret the intercepts or domain and range, but the focus of this lesson builds students' toolkits in graphing, which they can then apply in the context of real-world situations in the next lesson.

Component Summary				
	4.5.1 Warm-Up (~5 Min)	Counterexamples activity to facilitate critical thinking *(SE p. 184)*	4.5.4 Your Turn! (15 min)	Students practice graphing functions and interpreting key features (given a context and slope-intercept form) *(SE p. 188)*
	4.5.2 Guided Instruction (10 min)	Students explore key features leading to an interpretation of those features in the context *(SE p. 184)*	4.5.5 Wrap-Up (~10 min)	Students determine the error of key features that were identified and their interpretations *(SE p. 189)*
	4.5.3 Try It (10 min)	Graph and identify key features in context and function notation *(SE p. 186)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Intercepts • Slope • Domain, Range	• Calculators • (Optional) Straightedge for graphing	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle rewriting equations to slope-intercept form.
Students may attempt to add or subtract coefficients when solving (i.e., if $2x = 6$, they may subtract 2 on both sides rather than divide).
Students may reverse the context of the domain and the range in their interpretation.
Students may have trouble using function notation in 4.5.3 Try It, both in graphing and substituting values using function notation.</td><td>Students who took Pre-Algebra prior to Algebra 1 will have experience with slope, graphing, and basic interpretation. If students went into Algebra 1 without Pre-Algebra, this may be their first experience with these terms and skills.
Consider using On-Ramp video resources for additional scaffolding for students who need additional support in navigating the coordinate plane.
Allowing students to use calculators during the lesson will help students focus on the key features and their interpretations.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
During 4.5.1Warm-Up, students provide counterexamples to a given statement. Consider calling on students to explain, agree, disagree, or provide different counterexamples to give the opportunity for multiple pathways and mathematical discussions using a variety of student-created examples.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students connect key features with real-world contexts. Students look at linear relationships for different real-world contexts. Students graph the line to connect the equation to another representation. Students then connect the key features of the linear relationship to the real-world context. Students use the graph and the equation to interpret the key feature for the context.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider visual tools like GeoGebra or Desmos that can serve as learning companions during this lesson, particularly in 4.5.2 Guided Instruction, as students explore the meaning of domain and range in a given context. If resources permit, group students to digitally explore the line in Desmos before graphing on paper.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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4.5.1 Warm-Up: Counterexamples

Component Overview: *Students are provided with a list of statements to practice analyzing statements and producing counterexamples. Notice that the instructions include to only select three statements to increase student choice and entry points.*

4.5.1 Warm-Up: Counterexamples

A counterexample is a statement that shows that a general statement is false. For example, a counterexample for the general statement "All birds can fly" is "No, penguins are birds and they cannot fly."

1. Choose three of the statements and write a counterexample for each.

Statement	Counterexample
All prime numbers are odd.	*2 is a prime number*
Division of two values always results in a smaller value.	$4 \div \frac{1}{2} = 8$
The product of two numbers is greater than their sum.	$\frac{1}{2} \cdot 2 = 1$
All lines either increase from left to right or decrease from left to right.	*Horizontal or vertical line*
All lines have a y-intercept.	*Vertical lines*

To extend student thinking beyond the four walls of the classroom, consider adding a homework prompt for students to create a statement (that has a counterexample) to give to the class or a fellow student.

The challenge will be to construct the most creative and/or most challenging statement that will stump the class. Give a prize to whoever creates a statement that stumps the entire class!

Encourage students to choose what they feel confident with, or have students choose one that they consider to be 'easy' and one they consider to be 'challenging.'

4.5.2 Guided Instruction: Graphing a Line and Interpreting Key Features

Component Overview: *Students use multiple representations of a line to graph the line and to interpret its key features.*

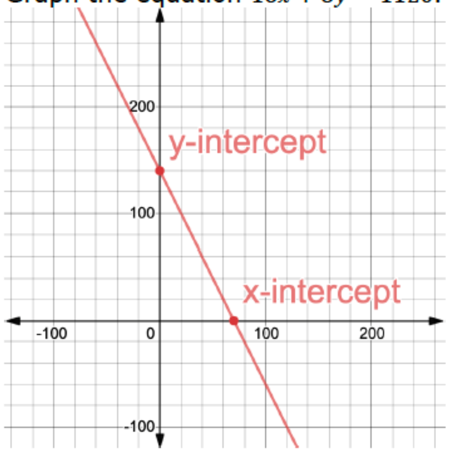
4.5.2 Guided Instruction: Graphing a Line and Interpreting Key Features

1. Lake Wales Little Theater is an intimate 140-seat theater. The theater uses the equation $16x + 8y = 1120$ to ensure each performance earns $\$1{,}120$ to cover the costs for a children's production. A table of values is shown.

Adult Tickets x	Child Tickets y	
0	140	*y-intercept*
18	104	
25	90	
56	28	
70	0	*x-intercept*

   a. Rewrite $16x + 8y = 1120$ in slope-intercept form.
   $8y = -16x + 1120$
   $y = -2x + 140$

   b. Graph the equation $16x + 8y = 1120$.



   c. Circle the x-intercept in the table and place a point on the graph.
   Answer on graph and table above.

Guided Instruction positions the teacher as facilitator, with students following prompts of intercepts to determine key features and interpret them in the context. Students first identify intercepts using different representations before interpreting their meaning in context, followed by a combination of both identifying and interpreting other key features in the same step.

Notice the scaffolding prompts for students to circle and label the intercepts both in the table of values and the graph. Students should make use of both the table of values and the graph throughout.

Is it possible to graph the equation or find the intercepts without converting to slope-intercept form? If so, how? Students can plot the intercepts and use them to find the slope. Consider recognizing students who might have seen this approach while rewriting to slope-intercept form.

4.5.2 Guided Instruction: Graphing a Line and Interpreting Key Features (continued)

d. Complete the interpretation of the x-intercept.
Lake Wales Little Theater will meet production costs

when it sells [○ 0 / ○ 70 / ○ 140] [○ adult / ○ child] tickets.

Answer: 70 adult

e. Circle the y-intercept in the table and place a point on the graph.
Answer on previous table and graph.

f. Complete the interpretation of the y-intercept.
Lake Wales Little Theater will meet production costs

when it sells [○ 0 / ○ 70 / ○ 140] [○ adult / ○ child] tickets.

Answer: 140 child

g. Complete the interpretation of the slope.
For every __2_____ child ticket(s) sold, Lake Wales Little Theater needs to sell __1_____ fewer adult ticket(s).

h. Complete the table that gives the other key features. Also write a brief explanation for each.

	Domain	Range	As x-values increase, y-values …
Key Feature	*0, 1, 2, 3…,70*	*0,1,2,3…,140*	*Decrease*
Reason	*The domain should be whole numbers up to 70 because a partial ticket could not be sold.*	*The range should be whole numbers up to 140 because a partial ticket could not be sold*	*The cost of an adult ticket is more so the more adult tickets that are sold, less child tickets are needed to make a profit.*

Students may interpret 70 adult tickets as "the" answer rather than one possible solution. Consider asking "If they don't sell 70 adult tickets, can they still meet production costs? How do you know?" in order for students to begin thinking about the answer in context in preparation for interpreting domain and range later in this activity.

Students should continue referencing both the table of values and the graph. Students should first identify slope using either the table of values or the graph. Avoid frontloading the information. Instead, provide students the space to recognize the need for finding slope and determining which method makes the most sense.

Consider which entry points in representing domain and range make the most sense for your students. They could spiral content from Lesson 1 in set-builder notation, use inequality symbols, or provide a description of the values.

4.5.3 Try It: Graphing and Interpreting Key Features

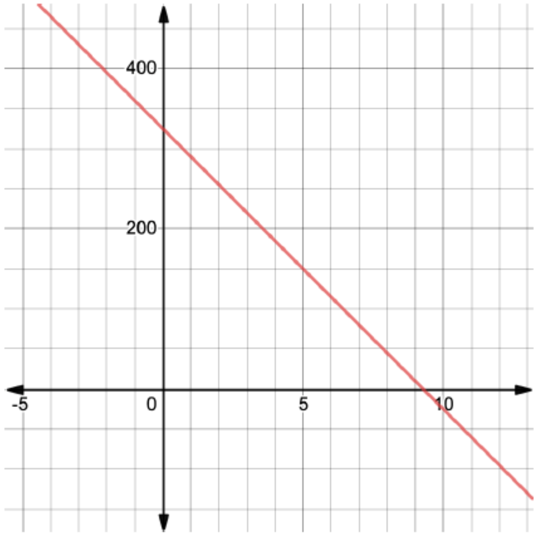
Component Overview: *Students will apply what they've learned about graphing linear functions and interpreting key features to a single problem.*

4.5.3 Try It: Graphing and Interpreting Key Features

The monarch butterfly is essential to the pollination process of many native and farmed North American plants. These butterflies migrate thousands of miles over several months.

A Florida Department of Forestry scientist has been tracking their population since the 2005 migration cycle. The scientist uses the function $B(x) = -35x + 325$ to estimate the number of butterflies, where x is the number of years since 2005 and B is the number of butterflies, in thousands.

1. Graph the function.



Notice the scaffolding has been removed compared to the previous activity. Students will need to interpret the context in order to label the axes without a table of values. This activity can be done independently or with a pair or group. The next component of this lesson is a Your Turn!. Consider allowing students to work together on these questions. Prior to releasing students to complete the task, review the word problem's context with the whole class. Once time or activity is complete, review all answers with the whole class.

Consider what scaffolding could look like for individual students or a small group in need of a lower entry point.

- *Provide a blank table of values and refer students back to the previous activity.*
- *Ask guiding questions regarding the axes in context. What does x represent? What does y represent? What does that mean for what the y-intercept means in context?*
- *Pair students with different strengths, for example, one who works well with graphs and one who is strong in verbally processing meaning in a given context.*

4.5.3 Try It: Graphing and Interpreting Key Features (continued)

2. Solve $-35x + 325 = 0$.
   $-35x = -325$
   $x = \frac{325}{35}$ or $x = \sim 9.286$

3. Which intercept is the solution to $-35x + 325 = 0$?
   x-intercept

4. Interpret the intercept.
   There are no monarch butterflies 9.286 *years after* 2005 *or sometime in* 2015

5. Find $B(0)$.
   $B(0) = -35(0) + 325$
   $= 325$

6. Which intercept is the value of $B(0)$?
   y-intercept

7. Interpret the intercept.
   In 2005 *there are* 325 *thousand monarch butterflies*

8. Complete the table.

	Domain	Range	As x-values increase, y-values …
Key Feature	$0 \le x \le 9.286$	$0 \le y \le 325$	*Decrease*
Reason	*Years there are butterflies*	*The range of the number of butterflies, in thousands, for the years in the domain*	*The population is getting smaller*

How can you determine which intercept is the solution and which is $B(0)$? What is the significance of each intercept in terms of the context of this problem?

Pay close attention to what students do in questions 3, 4, 6, and 7. Some students may be able to do the math to find the intercept but may be unable to interpret it within the context. Some students may have correct interpretations but have them attached to the wrong intercept. Correcting one while they are discussing can help correct the other.

Consider which entry points in representing domain and range make the most sense for your students. They could spiral content from Lesson 1 in set-builder notation, use inequality symbols, or provide a description of the values.

Take Turns
Students may be identifying the correct domain or range but interpreting differently. Consider having students share their rationale and interpretation with a partner to build mathematical discussion into this activity.

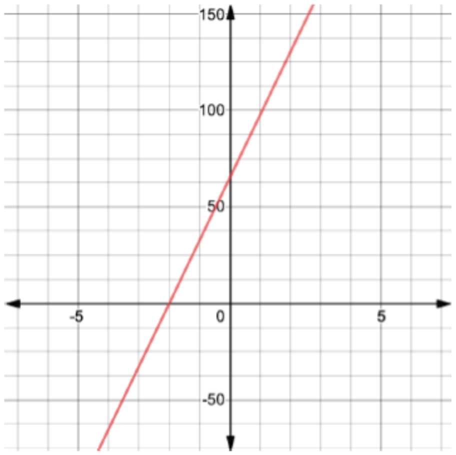
4.5.4 Your Turn!

Component Overview: Students continue practicing graphing functions and interpreting key features. This activity is designed to be done independently to give students an opportunity to try it and practice on their own.

4.5.4 Your Turn!

At Pensacola Fitness Gym, the cost for a gym membership is given by the equation $y = 32.50x + 65$, where y is the total cost in dollars for a membership for a certain number of months, x. The gym charges an initial membership fee and a monthly fee.

1. Graph the equation.



2. Determine the key features.

Domain	Range	As x-values increase, y-values ...
$0, 1, 2, 3, 4, \ldots$	$y \geq 65$	*increase*
Slope	**x-intercept**	**y-intercept**
32.5	-2	65

3. Interpret the y-intercept.
   The initial membership fee is $\$65.00$.

4. Interpret the slope.
   For each month the fee is $\$32.50$

Notice the shift from function notation to slope-intercept for the equation of a line. Any anchor charts or scaffolding on the graph of a line might be helpful here for students who would benefit from using manipulative tools for conceptualizing the key features in this context.

Students will be identifying a maximum value in the domain and range in the Wrap-Up. Consider asking here for students to determine whether there is a maximum for domain or range in this context and why.

4.5.5 Wrap-Up: Error Analysis

Component Overview: *Students will be given a set of key feature values and their interpretations for a function. They will apply their knowledge to analyze the values and interpretations to locate the error and fix it. This activity is designed to be done independently to provide data on their understanding of the content of this lesson.*

4.5.5 Wrap-Up: Error Analysis

The key features and the interpretations given for the context have errors. Find one of the errors and fix it.

1. The average price of a gallon of orange juice from October 2013 to September 2014 can be modeled by the function f shown, where x represents the number of months since October 2013.

$$f(x) = 6.12 + 0.03x$$

	Domain	Range	x-intercept
Key Feature	$x \geq 0$	$y \geq 6.12$	6.12

Interpretations	
Slope	**y-intercept**
For every gallon of orange juice, the month increased by 0.03.	In October 2013, the average price of a gallon of orange juice was $\$6.12$.

1. *The domain has a max value since the time is given as October 2013 to September 2014 which is 12 months. Domain is $0 \leq x \leq 12$. The domain does not need to be whole numbers since a fraction of a month is a possible time*
2. *Since there are 12 months the y-values also have a max. Range is $6.12 \leq y \leq 6.48$*
3. *6.12 is the y-intercept. The x-intercept is $x = -240$*
4. *The slope should reference that for every month it is the cost of the orange juice that increases by $\$0.03$.*

Consider all the possible skills or strands within this question that could be used to differentiate your support in Lesson 6.

- *Analyzing the thinking of others - students may only be able to identify the answers on their own and then compare those answers to the work provided, rather than engage in true error analysis as described in MA.K12.MTR.4.1.*
- *Interpretation of domain and range in context, particularly the concept of a minimum or maximum value in certain contexts*
- *Interpreting slope in relating change in y to change in x, which "order" should it be read in and why? (.03 is the increase in cost of orange juice, not the increase in month)*

Challenge students to find all four errors. Students who can correctly identify all four errors are able to interpret domain, range, intercepts, and slope in a context.